Supplementary Information

Bayesian Neural Surrogates Unify Uncertainty Propagation,
Sensitivity Attribution and Calibration for Coupled
Multi-Physics Reactor Analysis

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1 Supplementary Figures

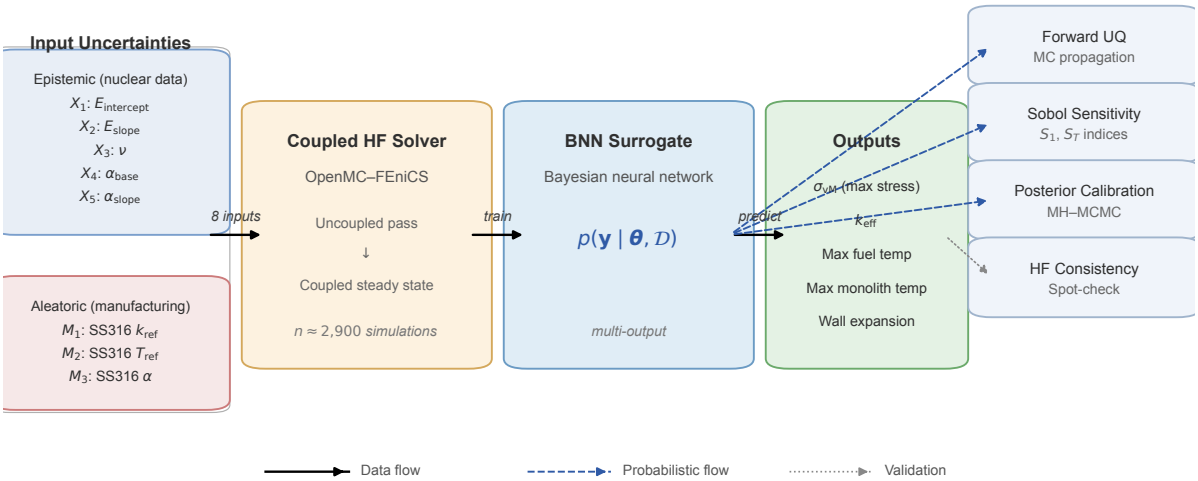

Figure 1: **Fig. S1 | BNN surrogate workflow.** Schematic of the end-to-end probabilistic analysis framework. The BNN surrogate is trained on coupled OpenMC–FEniCS data and produces a posterior predictive distribution. This single distribution serves as the shared probabilistic layer for forward uncertainty propagation, Sobol sensitivity attribution, and MCMC posterior calibration, without retraining between analyses.

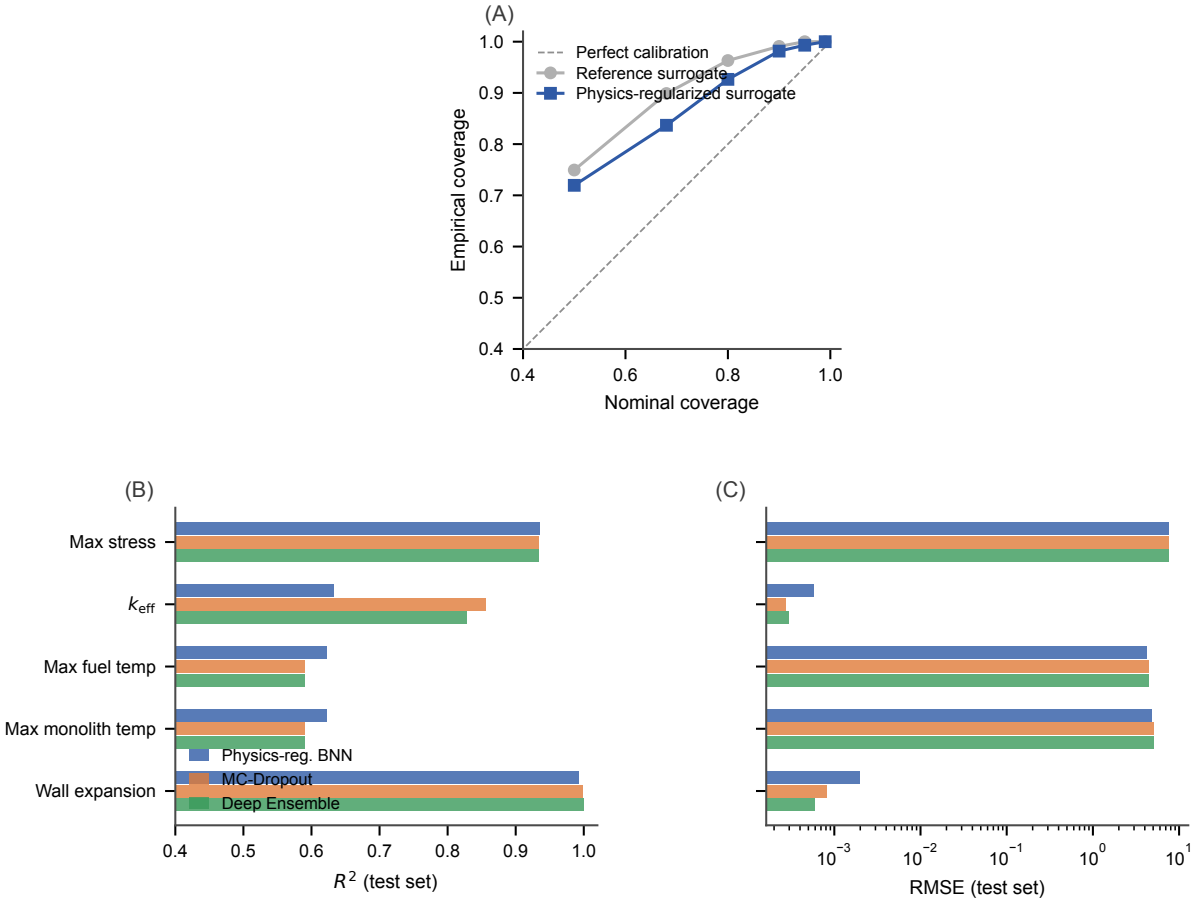

Figure 2: **Fig. S2 | Model validation: parity and residual analysis.** Predicted versus true values and residual distributions for primary outputs across both BNN variants (501 test samples). Shaded regions show the 90% predictive interval.

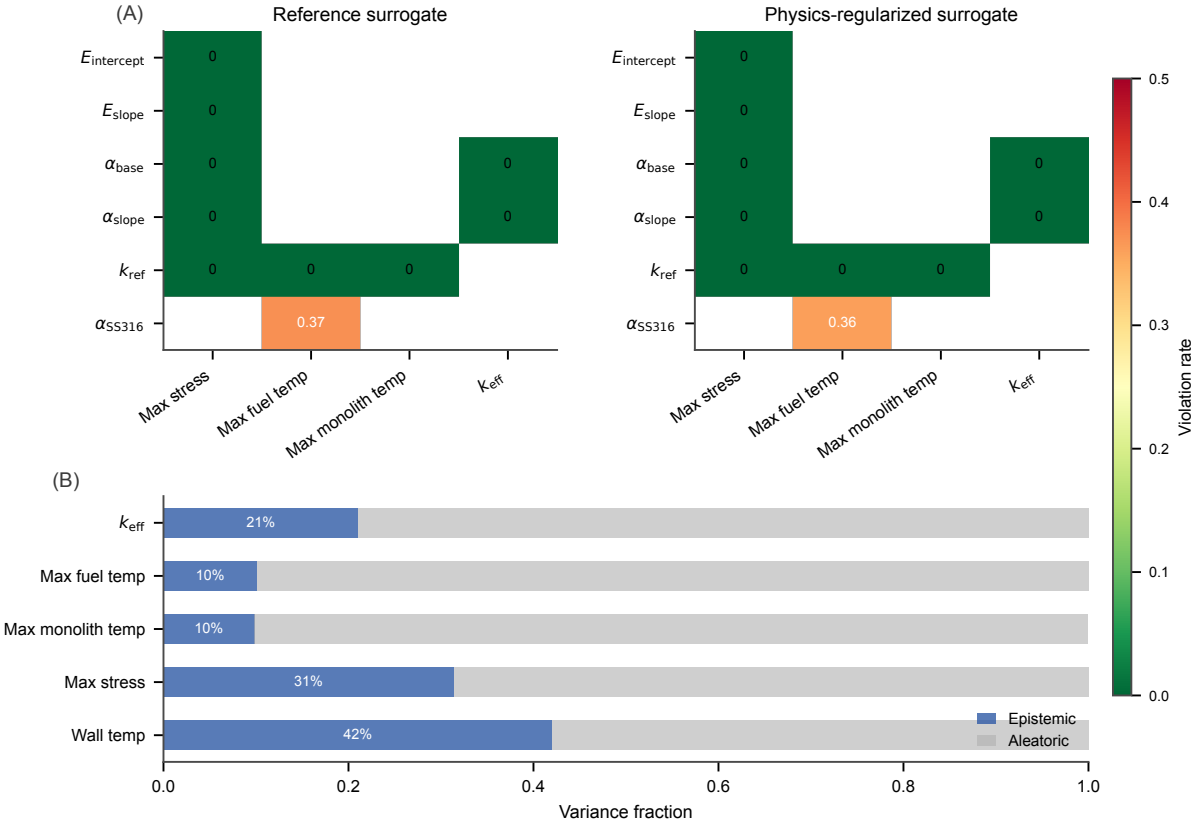

Figure 3: **Fig. S3 | Physics consistency and uncertainty decomposition.** (**A**) Monotonicity violation rates for 15 input–output pairs across both BNN variants; high-confidence pairs achieve zero violations. (**B**) Epistemic vs. aleatoric variance decomposition. Aleatoric uncertainty dominates across all outputs (58–90% of total variance).

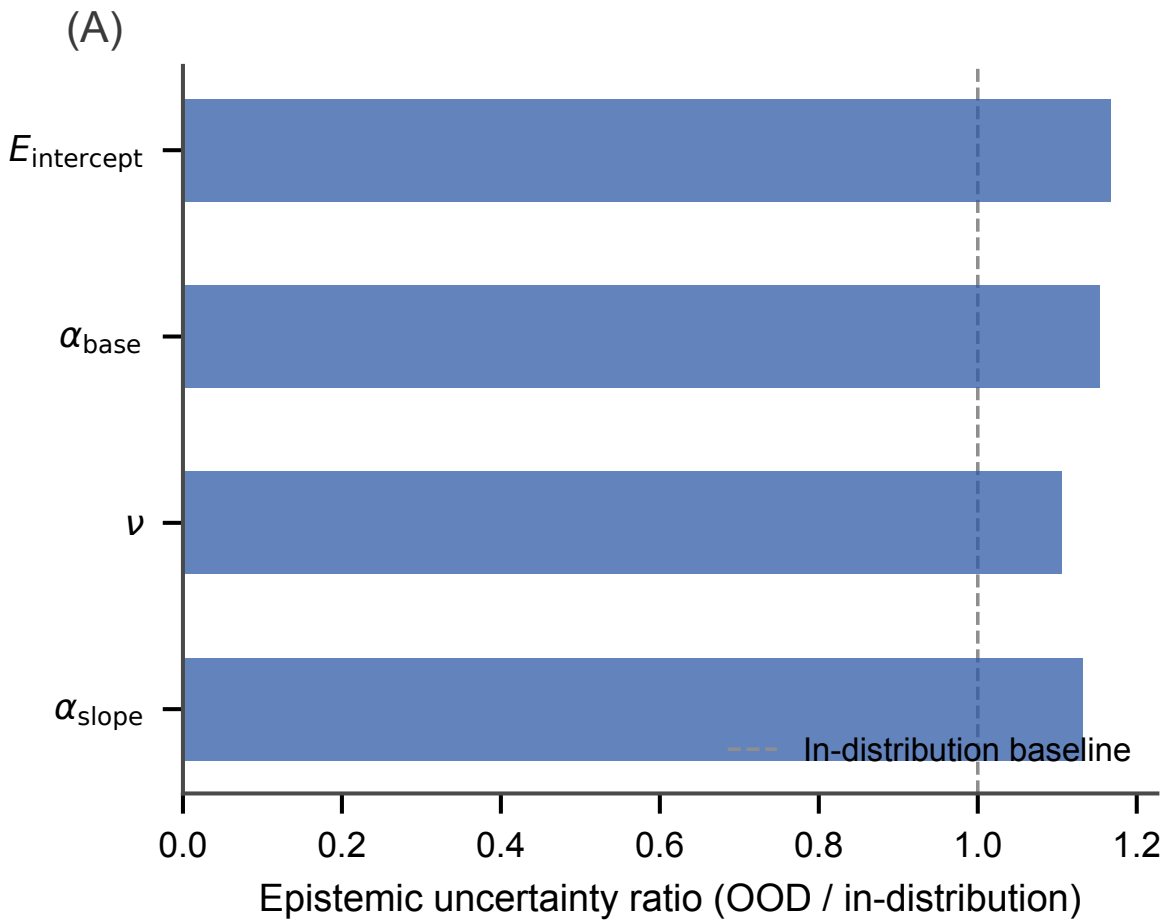

Figure 4: **Fig. S4 | Out-of-distribution (OOD) coverage and epistemic uncertainty.** 90% predictive interval coverage for in-distribution versus four feature-tail subsets. Stress retains coverage above 0.75 across all tails. k_{eff} coverage degrades substantially on the α_{base} tail (R^2 drops to approximately 0.29), indicating that the Sobol and posterior conclusions for k_{eff} are valid only within the training range of this dominant parameter. Elevated epistemic ratios on the α_{base} tail confirm that the BNN correctly signals increased model uncertainty outside the training domain.

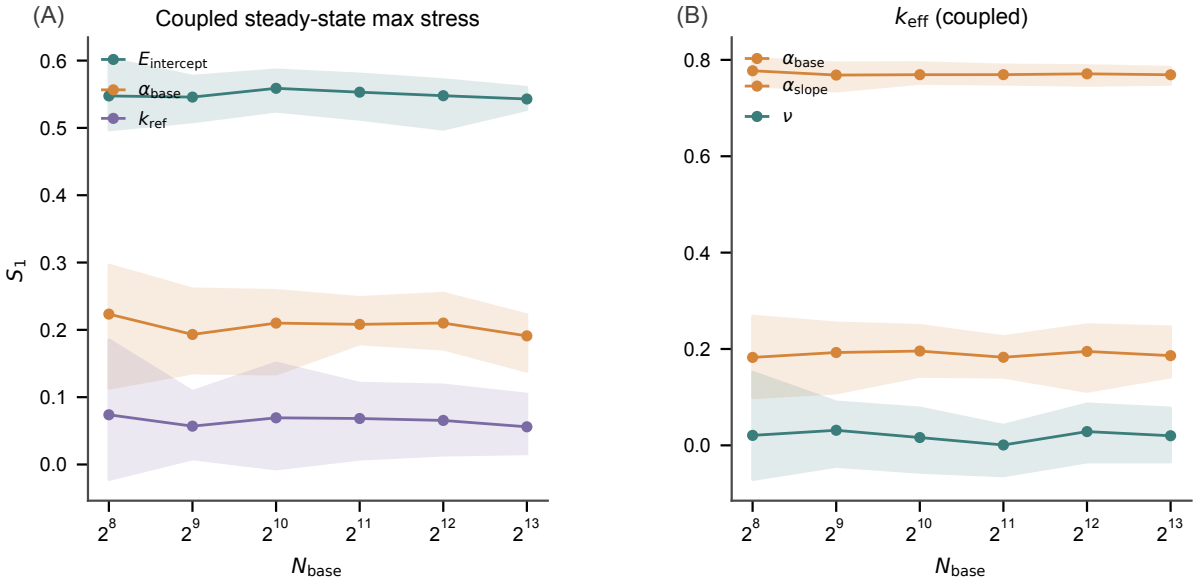

Figure 5: **Fig. S5 | Sobol index convergence with sample size.** First-order Sobol index S_1 as a function of sample size N_S (256 to 4096) for stress and k_{eff}, with 90% CI bands across 50 replications. Indices stabilise by $N_S = 2048$, confirming convergence at the chosen $N_S = 4096$.

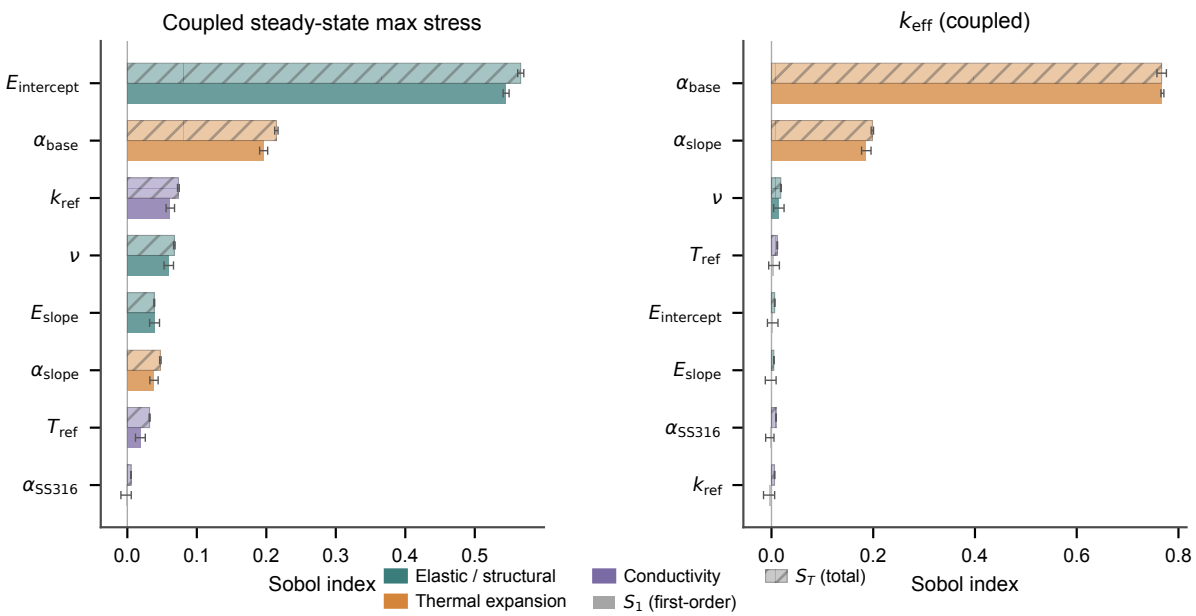

Figure 6: **Fig. S6 | Detailed Sobol sensitivity analysis.** First-order and total-order Sobol indices for all eight material parameters across all five primary outputs, showing the dominance of $E_{\text{intercept}}$ for stress and α_{base} for k_{eff}.

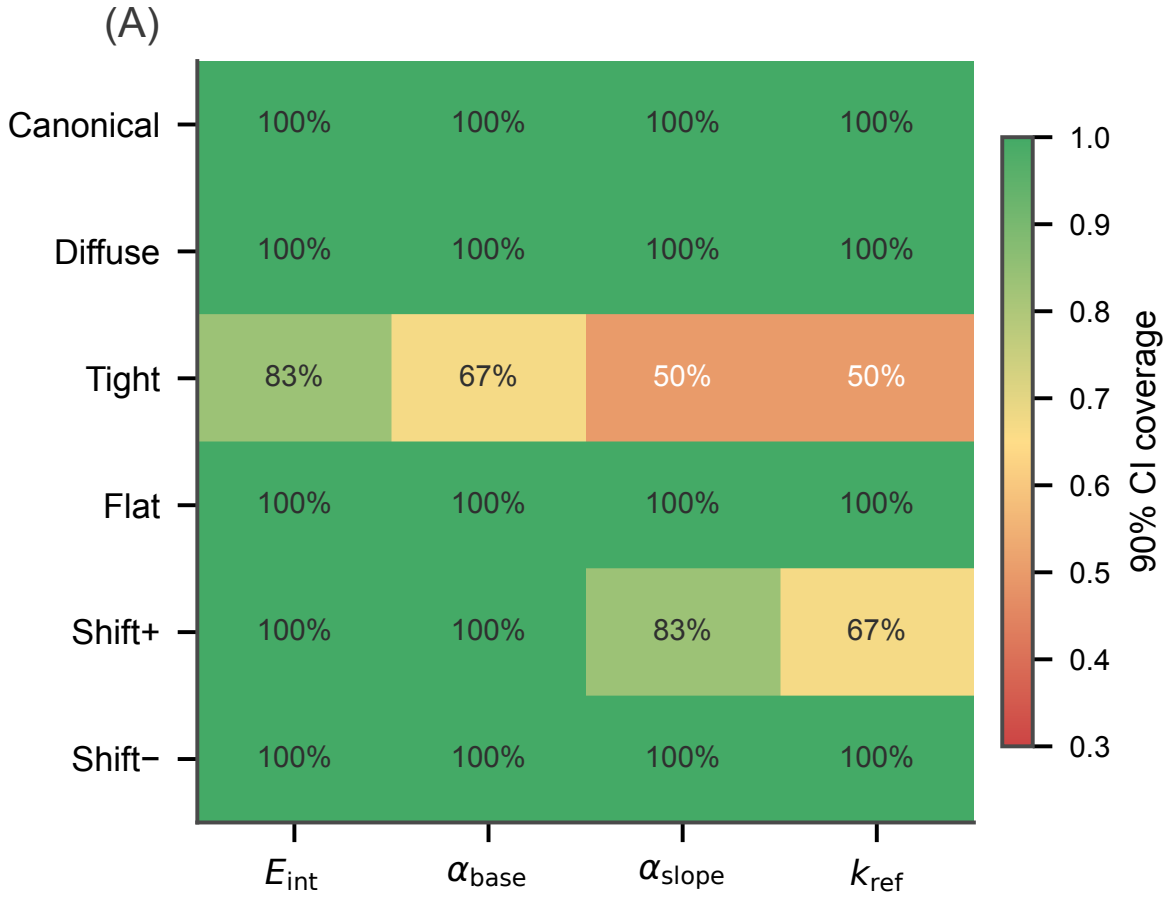

Figure 7: **Fig. S7 | Prior sensitivity analysis.** Posterior marginal distributions for the four calibrated parameters under six prior configurations: canonical, diffuse ($2\times$), tight ($0.5\times$), flat (uniform), shift$_+$ ($\mu + 0.5\sigma$), shift$_-$ ($\mu - 0.5\sigma$). The tight prior over-constrains the posterior, reducing coverage to 50–83% (Table S10).

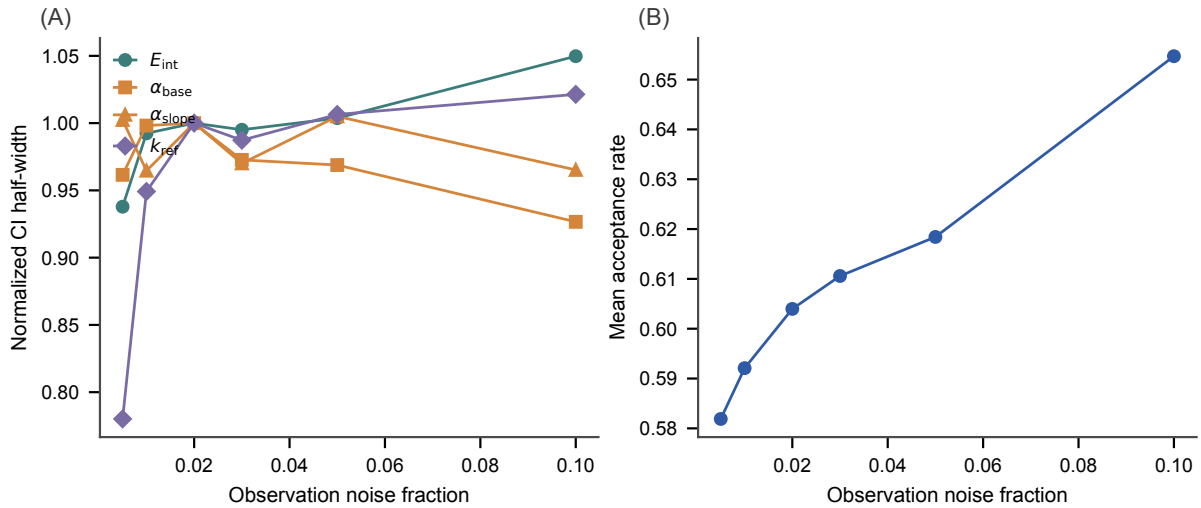

Figure 8: **Fig. S8 | Observation noise sensitivity.** Posterior width and acceptance rate as a function of assumed observation noise fraction (0.5%–10%). Posterior standard deviations increase monotonically; acceptance rates increase appropriately as the likelihood broadens (Table S11).

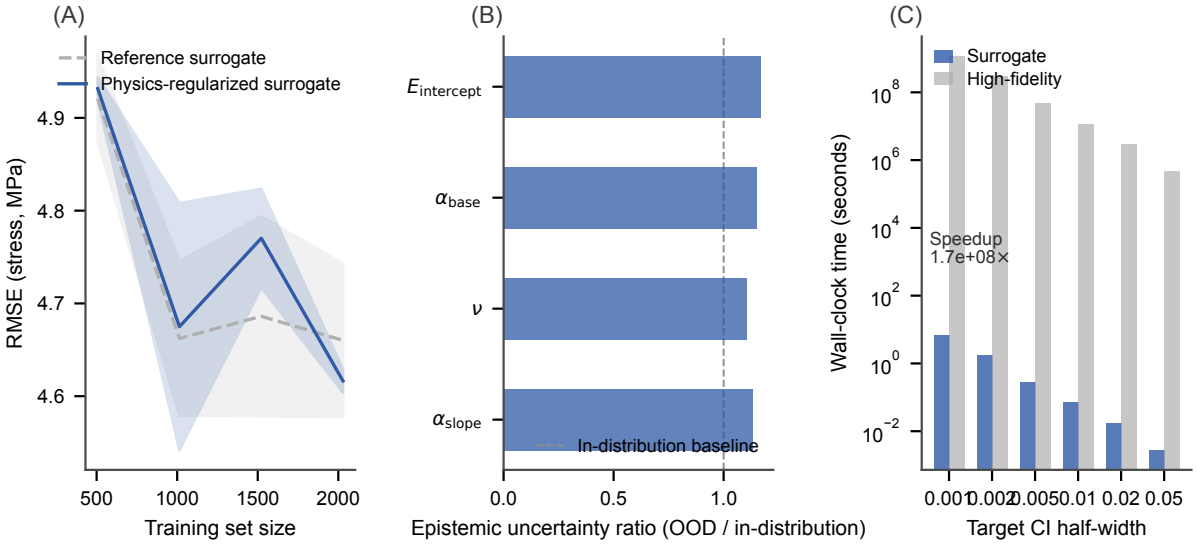

Figure 9: **Fig. S9 | Data efficiency and computational cost.** Test RMSE and R^2 as a function of training set size for the physics-regularized BNN. Near-saturated performance is achieved at 25% data utilisation (507 training samples; Table S12).

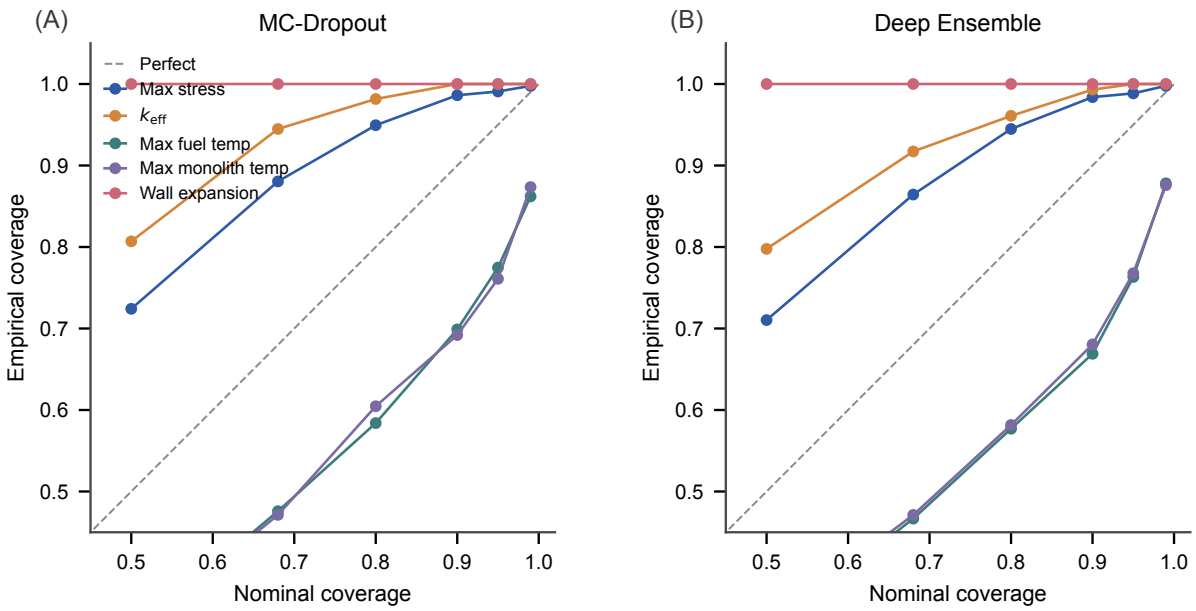

Figure 10: **Fig. S10 | External baseline calibration comparison.** Calibration curves comparing BNN, MC-Dropout and deep ensemble across primary outputs. The BNN achieves substantially better probabilistic calibration (ECE 0.027 vs. 0.18–0.19) on temperature outputs.

2 Supplementary Tables

Table 1: **Table S1 | Material parameters: definitions, nominal values and priors.**

Parameter	Symbol	Nominal value	Unit	Prior range ($\pm$10%)
Young's mod. intercept	$E_{\text{intercept}}$	2×10^{11}	Pa	$[1.8, 2.2] \times 10^{11}$
Young's mod. slope	E_{slope}	-7×10^{7}	Pa/K	$[-7.7, -6.3] \times 10^{7}$
Poisson's ratio	ν	0.31	—	[0.279, 0.341]
CTE base	α_{base}	1×10^{-5}	K^{-1}	$[0.9, 1.1] \times 10^{-5}$
CTE slope	α_{slope}	5×10^{-9}	K^{-2}	$[4.5, 5.5] \times 10^{-9}$
Ref. conductivity	k_{ref}	23.2	W/(m·K)	[20.88, 25.52]
Ref. temperature	T_{ref}	923.15	K	[830.84, 1015.47]
Cond. slope	k_{slope}	1/75	W/(m·K^2)	[0.012, 0.0147]

Table 2: **Table S2 | BNN hyperparameter search space and optimal values.**

Hyperparameter	Search range	Optimal (bnn-phy-mono)
Width	[64, 256]	254
Depth	[2, 8]	2
Learning rate	$[10^{-4}, 10^{-2}]$	9.81×10^{-4}
Weight decay	$[10^{-9}, 10^{-5}]$	8.09×10^{-8}
Batch size	{16, 32, 64}	32
Epochs	[200, 500]	399
Gradient clip	[0.1, 2.0]	0.674
w_{data}	[0.1, 5.0]	1.855
Prior σ	[0.1, 2.0]	0.478
KL weight	$[10^{-5}, 10^{-2}]$	1.93×10^{-4}
w_{mono}	[0.01, 1.0]	0.212
$\rho_{\text{abs,min}}$	[0.01, 0.5]	0.123
mono_topk	[10, 100]	53

Search: Optuna with 200 trials; objective: validation RMSE. Activation: SiLU throughout. Output heads: mean + log-variance (heteroscedastic).

Table 3: **Table S3 | BNN ablation study: reference baseline vs. physics-regularized surrogate** (mean $\pm$ s.d., 5 seeds). Additional ablations (heteroscedastic noise, multi-fidelity architecture) are in Notes G–I.

Model variant	Stress R^2	Stress PICP	k_{eff} R^2
Reference (ELBO only)	0.936 ± 0.007	0.995 ± 0.002	0.633 ± 0.260
Physics-regularized	0.935 ± 0.007	0.987 ± 0.007	0.633 ± 0.258

Table 4: **Table S4 | Complete Sobol indices (S_1 and S_T) for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI).

Parameter	Stress S_1 [90% CI]	Stress S_T [90% CI]
E_{slope}	0.037 [−0.046, 0.100]	0.039 [0.038, 0.039]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	0.566 [0.562, 0.571]
ν	0.063 [0.034, 0.097]	0.068 [0.067, 0.069]
α_{base}	0.210 [0.171, 0.254]	0.214 [0.212, 0.217]
α_{slope}	0.038 [−0.013, 0.091]	0.047 [0.046, 0.049]
T_{ref}	0.019 [−0.025, 0.055]	0.032 [0.031, 0.033]
k_{ref}	0.065 [0.014, 0.118]	0.073 [0.072, 0.075]
SS316_α	−0.004 [−0.068, 0.034]	0.005 [0.005, 0.006]
Parameter	k_{eff} S_1 [90% CI]	k_{eff} S_T [90% CI]
E_{slope}	0.010 [−0.052, 0.062]	0.005 [0.004, 0.005]
$E_{\text{intercept}}$	0.008 [−0.065, 0.071]	0.006 [0.006, 0.007]
ν	0.029 [−0.035, 0.086]	0.018 [0.018, 0.019]
α_{base}	0.771 [0.747, 0.788]	0.767 [0.758, 0.776]
α_{slope}	0.195 [0.112, 0.250]	0.198 [0.196, 0.201]
T_{ref}	0.010 [−0.023, 0.054]	0.011 [0.011, 0.012]
k_{ref}	0.000 [−0.063, 0.048]	0.006 [0.006, 0.007]
SS316_α	0.003 [−0.059, 0.074]	0.009 [0.009, 0.009]

Table 5: **Table S5 | Posterior calibration: 18 benchmark case summary** (physics-regularized BNN).

Case	Category	True stress (MPa)	Accept rate	90%-CI coverage	Post-mean err.
0	low	110.3	0.607	3/4	8.3
1	low	117.7	0.605	4/4	24.6
2	low	98.7	0.632	1/4	9.0
3	low	115.9	0.623	3/4	8.8
4	low	96.8	0.612	2/4	3.1
5	low	102.0	0.617	3/4	23.4
6	near	124.5	0.609	4/4	20.3
7	near	121.4	0.592	4/4	26.6
8	near	127.2	0.616	4/4	5.9
9	near	129.4	0.602	4/4	3.5
10	near	122.4	0.597	4/4	4.5
11	near	130.2	0.595	4/4	3.8
12	high	135.0	0.608	4/4	0.2
13	high	148.5	0.594	2/4	12.1
14	high	141.2	0.582	4/4	13.7
15	high	198.4	0.595	4/4	3.6
16	high	133.2	0.622	4/4	10.5
17	high	188.6	0.593	4/4	5.4
Overall			0.58–0.63	62/72 = 0.861	MAE = 10.4

Table 6: **Table S6 | Computational cost breakdown.**

Operation	Time	Platform
Coupled HF solve (1 case)	2266 s	EPYC 9654 / 96 cores
BNN single draw ($M = 50$)	1.29×10^{-2} s	GPU
BNN batched (batch=1024, per sample)	1.34×10^{-5} s	GPU
20,000-sample forward UQ	$\sim$0.3 s	GPU
Sobol (50 reps $\times$ 40,960 evals)	$\sim$minutes	GPU
18-case MCMC (8000 iter $\times$ 4 chains)	$\sim$minutes	GPU
Speedup (single draw vs. HF)	1.43×10^5	
Speedup (batched vs. HF)	1.37×10^8	

Table 7: **Table S7 | Physics consistency: monotonicity violation rates** (501 test samples, discrete $\pm\delta$ perturbation test).

Physics pair (input $\rightarrow$ output, sign)	Ref. BNN	Phy-reg
$E_{\text{intercept}} \rightarrow$ Max stress (+)	0%	0%
$\alpha_{\text{base}} \rightarrow$ Max stress (+)	0%	0%
$\alpha_{\text{base}} \rightarrow k_{\text{eff}}$ ($-$)	0%	0%
$k_{\text{ref}} \rightarrow$ Max stress ($-$)	0%	0%
SS316_$\alpha \rightarrow$ Fuel temp ($-$)	36.3%	35.6%

Ordering/non-negativity constraints (max $\geq$ average temp, stress ≥ 0): 0% violations for both BNN variants.

Table 8: **Table S8 | Epistemic vs. aleatoric uncertainty decomposition on test set.**

	Ref. BNN		Phy-reg BNN	
Output	Epi%	Ale%	Epi%	Ale%
k_{eff}	12.0	88.0	10.2	89.8
Max fuel temp	12.0	88.0	10.1	89.9
Max monolith temp	11.2	88.8	10.1	89.9
Max stress	27.7	72.3	29.6	70.4
Wall expansion	34.1	65.9	41.7	58.3

Table 9: **Table S9 | External baseline: full comparison of scoring rules.**

Model	Output	RMSE	R^2	CRPS	NLL	ECE
MC-Dropout	k_{eff}	0.0003	0.856	0.0002	-6.502	0.152
	Fuel temp	4.497	0.590	2.624	3.352	0.183
	Monolith T	5.061	0.590	2.951	3.472	0.182
	Max stress	7.659	0.934	4.519	3.592	0.118
	Wall exp.	0.0008	0.999	0.002	-3.831	0.197
Deep Ensemble	k_{eff}	0.0003	0.828	0.0002	-6.476	0.142
	Fuel temp	4.495	0.590	2.634	3.326	0.190
	Monolith T	5.061	0.590	2.962	3.439	0.185
	Max stress	7.644	0.934	4.502	3.586	0.112
	Wall exp.	0.0006	0.999	0.002	-3.841	0.197

Table 10: **Table S10 | Prior sensitivity** (6 configurations, physics-regularized BNN, case 06, near-threshold).

Prior variant	Coverage (4 params)	Mean KL vs. canonical	Accept rate
Canonical	100%	0.00 (reference)	0.61
Diffuse ($2\times$)	100%	0.15–0.22	0.63
Tight ($0.5\times$)	50–83%	0.64–1.28	0.57
Flat (uniform)	100%	0.24–0.38	0.64
Shift ($+0.5\sigma$)	67–100%	0.06–0.36	0.60
Shift (-0.5σ)	83–100%	0.08–0.28	0.61

Table 11: **Table S11 | Noise sensitivity** (noise fraction 0.5%–10%, physics-regularized BNN, case 06).

Noise fraction	$E_{\text{intercept}}$ post. std	Accept rate	Coverage
0.5%	1.26×10^{10}	0.58	100%
1.0%	1.28×10^{10}	0.59	100%
2.0% (default)	1.32×10^{10}	0.61	100%
5.0%	1.36×10^{10}	0.63	100%
10.0%	1.40×10^{10}	0.65	83%

3 Supplementary Notes

3.1 Note A: BNN training details

A.1. Evidence lower bound (ELBO) derivation

The BNN posterior over weights $\mathbf{w}$ is approximated by a mean-field variational distribution $q(\mathbf{w}) = \prod_i \mathcal{N}(w_i \mid \mu_i, \sigma_i^2)$. The variational parameters are optimised by minimising the negative evidence lower bound:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathrm{KL}\big[q(\mathbf{w}) \,\|\, p(\mathbf{w})\big] - \mathbb{E}_{q(\mathbf{w})}\big[\log p(\mathcal{D} \mid \mathbf{w})\big] \tag{1}$$

where $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$ is the isotropic Gaussian prior. The KL divergence admits a closed form; the data-fit term is approximated by a single Monte Carlo sample per mini-batch (Bayes-by-Backprop[10]).

The reparameterisation trick enables gradient computation: $w = \mu + \sigma \cdot \epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. The variance parameter $\sigma = \log(1 + \exp(\rho))$ ensures positivity, with $\rho_{\text{abs,min}}$ clamping $\rho \geq \rho_{\text{abs,min}}$ to prevent posterior collapse.

A.2. Heteroscedastic observation noise

The network outputs both a mean $\mu(\boldsymbol{\theta})$ and a log-variance $\log \sigma^2_{\text{obs}}(\boldsymbol{\theta})$ per output channel:

$$\log p(y_j \mid \boldsymbol{\theta}, \mathbf{w}) = -\tfrac{1}{2}\big[\log \sigma^2_{\text{obs},j} + (y_j - \mu_j)^2 / \sigma^2_{\text{obs},j}\big] \tag{2}$$

This heteroscedastic formulation allows the BNN to model input-dependent observation noise, separating it from epistemic (weight) uncertainty.

A.3. KL annealing schedule

The KL weight β is annealed from 0 to its target value kl_weight $= 1.93 \times 10^{-4}$ over the first 100 epochs: $\beta(t) = \min(t/100, 1) \cdot$ kl_weight. This prevents the KL term from dominating early training before the data-fit term has found a reasonable region of weight space.

A.4. Physics-prior monotonicity constraints

The physics-regularized BNN adds an auxiliary loss:

$$\mathcal{L}_{\text{mono}} = \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\ -\text{sign} \cdot \partial\mu / \partial\theta_k\big) \tag{3}$$

weighted by $w_{\text{mono}} = 0.212$. Sign constraints are derived from constitutive physics—for example, stress increases with $E_{\text{intercept}}$ because $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$ and $\mathbf{C}$ is linear in E.

These are physics-prior constraints derived from constitutive relationships.

A.5. Gradient clipping and initialisation

Gradient norms are clipped at 0.674 per parameter group. Weight means μ are initialised with Xavier uniform; ρ values are initialised uniformly in $[-5, -4]$, corresponding to small initial variance. These choices stabilise early training when the KL term is annealed in.

3.2 Note B: Sobol sensitivity methodology

B.1. Sobol variance decomposition

For a square-integrable function $f(\theta_1, \ldots, \theta_d)$ with independent inputs, the total variance $V = \mathrm{Var}[f]$ can be decomposed as:

$$V = \sum_i V_i + \sum_{i<j} V_{ij} + \cdots + V_{1,\ldots,d} \tag{4}$$

The first-order Sobol index is $S_i = V_i/V$; the total-order index is $S_{T,i} = 1 - V_{\sim i}/V$.

B.2. Saltelli cross-matrix scheme

Two independent $N \times d$ matrices $\mathbf{A}$ and $\mathbf{B}$ are drawn from the joint prior. For each input i, a cross-matrix $\mathbf{A}_{\mathbf{B}}^{(i)}$ is constructed by replacing the i-th column of $\mathbf{A}$ with the i-th column of $\mathbf{B}$. The model is evaluated at all $N(d+2)$ points.

B.3. Jansen estimator

$$S_{1,i} = 1 - \frac{1}{2N} \sum_{j=1}^{N} \big[f(\mathbf{A}_{\mathbf{B}}^{(i)})_j - f(\mathbf{B})_j \big]^2 / V \tag{5}$$

$$S_{T,i} = \frac{1}{2N} \sum_{j=1}^{N} \big[f(\mathbf{A})_j - f(\mathbf{A}_{\mathbf{B}}^{(i)})_j \big]^2 / V \tag{6}$$

The Jansen estimator is preferred for its lower variance compared to the Sobol estimator, particularly for indices near zero.

B.4. Replication-based confidence intervals

The entire Sobol analysis (sample generation, model evaluation, index estimation) is independently repeated $R = 50$ times. The 90% confidence interval for each index is constructed from the 5th and 95th percentiles of the R replications. A factor is declared a stable contributor only if the lower bound of its 90% CI exceeds zero.

The BNN posterior predictive mean (averaged over $M = 30$ weight samples) serves as the deterministic function $f(\boldsymbol{\theta})$ in the Sobol framework.

3.3 Note C: MCMC posterior calibration details

C.1. Likelihood formulation

$$\log p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \sum_{j=1}^{J} \log \mathcal{N}\big(y_{\text{obs},j} \mid \mu_j(\boldsymbol{\theta}),\ \sigma^2_{\text{obs},j} + \sigma^2_{\text{BNN},j}(\boldsymbol{\theta})\big) \tag{7}$$

where $\mu_j(\boldsymbol{\theta})$ is the BNN posterior predictive mean (averaged over $M = 20$ weight samples), $\sigma^2_{\text{BNN},j}$ is the BNN aleatoric variance, and $\sigma^2_{\text{obs},j}$ is the observation noise variance (2% of the true value). The $J = 5$ primary outputs are: coupled peak global stress, coupled peak fuel temperature, coupled peak monolith temperature, coupled wall expansion, and k_{eff}.

C.2. Proposal distribution

A symmetric Gaussian random-walk proposal is used: $\boldsymbol{\theta}^* = \boldsymbol{\theta}_{\text{current}} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{proposal}})$, where $\boldsymbol{\Sigma}_{\text{proposal}} = \text{diag}(\text{scale}_1^2, \ldots, \text{scale}_4^2)$. The proposal scales were calibrated by short pilot runs of 1000 iterations.

C.3. Chain configuration

For each benchmark case:

- 4 independent chains with dispersed initialisations drawn from the prior
- $N_{\text{total}} = 8000$ iterations per chain
- Burn-in: first 2000 iterations discarded
- Thinning: every 5th post-burn-in sample retained
- Effective posterior: 1200 samples per chain, 4800 total per case

The four calibrated parameters are $E_{\text{intercept}}$, α_{base}, α_{slope}, and k_{ref}, selected as the four with the largest Sobol total-order indices across stress and k_{eff}. The remaining four parameters (E_{slope}, ν, T_{ref}, k_{slope}) are fixed at their true values.

C.4. Convergence diagnostics

Convergence is assessed by:

1. Visual inspection of trace plots (Fig. S9 in the main text references)
2. Rank-normalised split-$\hat{R}$ (Vehtari et al. 2021): $\hat{R}_{\text{max}} = 1.017$ across all 72 parameter–case combinations, well below the 1.1 threshold
3. Geyer initial-positive-sequence effective sample size: $\text{ESS}_{\text{min}} = 352$ (all cases)
4. Acceptance rates: range 0.58–0.62, mean 0.605 (Table S5)

C.5. Benchmark case selection

The 18 benchmark cases are drawn from the 501-sample test split:

- 6 low-stress cases (true stress 102–119 MPa)
- 6 near-threshold cases (true stress 123–129 MPa)
- 6 high-stress cases (true stress 143–198 MPa)

The threshold for categorisation is the primary safety threshold of 131 MPa. For each case, the "observation" is the true HF output vector perturbed by 2% multiplicative Gaussian noise.

3.4 Note D: Forward UQ extensions

D.1. Posterior predictive mean vs. full predictive draw

The main-text forward UQ propagates through the BNN posterior predictive mean $y = \mathbb{E}_{\mathbf{w}}[\mu_{\mathbf{w}}(\boldsymbol{\theta})]$, isolating parametric uncertainty from the BNN's own epistemic and aleatoric spread. As a consistency check, full predictive draws (sampling both weight uncertainty and observation noise) produce wider distributions but consistent means and percentile rankings, confirming that the parametric sensitivity structure is robust to the choice of predictive protocol.

D.2. Latin hypercube sampling

Latin hypercube sampling (LHS) is used rather than simple random sampling because LHS ensures more uniform coverage of the input space, reducing sampling variance for a given sample size. With $N = 20{,}000$ samples and $d = 8$ inputs, the LHS design partitions each input dimension into 20,000 equal-probability intervals.

3.5 Note E: Coupled simulation and material model details

E.1. Material constitutive models

The SS316 monolith is the structural component that simultaneously bears thermal stress and transmits heat between fuel and heat pipes. All eight uncertain parameters enter the calculation through the constitutive models:

- **Structural mechanics:** $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$; ν (Poisson's ratio). Stress: $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$.
- **Thermal deformation:** $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$. Dual role: thermal strain $\epsilon_{\text{th}} = \alpha \cdot \Delta T$ and geometric expansion.
- **Thermal transport:** $k(T) = k_{\text{ref}} + k_{\text{slope}} \cdot (T - T_{\text{ref}})$.

E.2. Picard iteration and convergence

The OpenMC–FEniCS coupling proceeds as:

1. Initialise with the eight material parameters and reference geometry.
2. OpenMC computes k_{eff} and the spatially resolved fission power distribution.
3. FEniCS solves for the temperature field, thermal expansion displacement and stress field.
4. The updated geometry and temperature-dependent material cross-sections are fed back to OpenMC.
5. Steps 2–4 repeat until convergence.

Convergence criterion: maximum temperature difference between consecutive iterations $< 1\,\text{K}$. Under-relaxation factor: 0.1.

E.3. Output definitions

Each HF evaluation produces 15 scalar outputs: 7 decoupled (before coupling convergence) and 8 coupled (after full convergence, including k_{eff}). The posterior likelihood uses 5 primary outputs: k_{eff}, peak fuel

temperature, peak monolith temperature, peak global stress, and wall expansion.

E.4. Dataset generation

$n = 3418$ samples were generated by Latin hypercube sampling over the 8-dimensional prior. The split (seed $= 2026$): training 2339 ($\sim$70%), validation 501 ($\sim$15%), test 501 ($\sim$15%). The fixed split is reused across all experiments for reproducibility.

3.6 Note F: High-fidelity coupled simulation framework

This note provides the governing equations of the four solver modules that generate the coupled training data consumed by the BNN surrogate. The framework couples OpenMC (neutron transport), FEniCS (thermo-elasticity), a one-dimensional heat-pipe thermal–hydraulic model and a fuel-temperature module via Picard fixed-point iteration. Details follow the implementation described in Refs. [9, 11].

F.1. Neutron transport — OpenMC

OpenMC [11] is an open-source Monte Carlo particle transport code supporting fixed-source, k-eigenvalue and subcritical multiplication calculations. In the coupled framework it computes k_{eff}, the fission power distribution and auxiliary neutronic tallies.

Integral transport equation. The steady-state, continuous-energy neutron transport equation in integral form reads

$$\Phi(\mathbf{r}, \mathbf{\Omega}, E) = \int_0^\infty e^{-\tau(\mathbf{r}, \mathbf{r} - s\mathbf{\Omega}, E)}\, Q(\mathbf{r} - s\mathbf{\Omega}, \mathbf{\Omega}, E)\, \mathrm{d}s \tag{8}$$

where Φ is the angular neutron flux at position $\mathbf{r}$, direction $\mathbf{\Omega}$ and energy E; Q is the emission density (scattering + fission source); and τ is the optical thickness:

$$\tau(\mathbf{r}, \mathbf{r}', E) = \int_0^{|\mathbf{r} - \mathbf{r}'|} \Sigma_t(\mathbf{r} - s\mathbf{\Omega}, E)\, \mathrm{d}s \tag{9}$$

with Σ_t the macroscopic total cross-section.

Emission density and Fredholm formulation. Substituting the scattering and fission kernels into the emission density yields the second-kind Fredholm integral equation:

$$Q(\mathbf{r}, \mathbf{\Omega}, E) = \int K(\mathbf{r}', \mathbf{\Omega}', E' \to \mathbf{r}, \mathbf{\Omega}, E)\, Q(\mathbf{r}', \mathbf{\Omega}', E')\, \mathrm{d}\mathbf{r}'\, \mathrm{d}\mathbf{\Omega}'\, \mathrm{d}E' \tag{10}$$

where the integral kernel K is the product of a transport kernel (free-flight attenuation) and a collision kernel (scattering + fission). Writing this in operator form $Q = \mathcal{K}Q$ and iterating:

$$Q^{(n+1)} = \mathcal{K}\, Q^{(n)}, \qquad Q = \lim_{n\to\infty} Q^{(n)} = \sum_{k=0}^{\infty} \mathcal{K}^k\, Q^{(0)} \tag{11}$$

which is the Neumann series solution. The converged Q is substituted back into Eq. (8) to recover the full neutron flux distribution.

Monte Carlo k-eigenvalue calculation. In practice, the integral equation is solved stochastically: a population of N_p neutron histories is tracked through successive fission generations. Each generation produces an estimate of

$$k_{\text{eff}} = \frac{\text{neutron production in generation } n+1}{\text{neutron production in generation } n} \tag{12}$$

and the converged fission source gives the spatial power distribution. The statistical uncertainty on k_{eff} is kept below 10 pcm (1σ) and the maximum power-distribution standard deviation below 0.19 % by using a sufficiently large number of particle histories per batch.

Geometry and temperature feedback. Thanks to the constructive solid geometry representation in OpenMC, the framework updates fuel-rod and heat-pipe coordinates (reflecting thermal expansion), material temperatures (Doppler broadening) and material densities (thermal expansion) between coupling iterations.

F.2. Thermo-elastic analysis — FEniCS

The thermo-elastic module is built on the open-source finite-element platform FEniCS [12]. It solves two coupled problems on the monolith mesh: steady-state heat conduction and linear thermoelasticity.

Heat conduction. The steady-state heat equation on the monolith domain Ω is

$$-\nabla \cdot \big[k(T)\,\nabla T\big] = \dot{q} \qquad \text{in } \Omega \tag{13}$$

where $k(T)$ is the temperature-dependent thermal conductivity of SS316 (see Note E) and $\dot{q}$ is the volumetric heat generation rate from fission. Boundary conditions are: isothermal walls at heat-pipe surfaces (temperatures from the heat-pipe module) and prescribed heat fluxes at fuel-rod surfaces (from the power distribution).

Linear thermoelasticity. For an isotropic material, the three-dimensional linear elastic constitutive relation (generalised Hooke's law) is

$$\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\,\mathbf{I} + 2\mu\,\boldsymbol{\varepsilon} \tag{14}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\boldsymbol{\varepsilon}$ the strain tensor, $\mathbf{I}$ the identity tensor, and λ, μ are the Lamé constants determined by the temperature-dependent Young's modulus $E(T)$ and Poisson's ratio ν:

$$\lambda = \frac{E\,\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)} \tag{15}$$

Thermal strain (Duhamel–Neumann law). Including thermal deformation, the constitutive equation for an isotropic solid becomes

$$\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\,\mathbf{I} + 2\mu\,\boldsymbol{\varepsilon} - (3\lambda + 2\mu)\,\alpha(T)\,\Delta T\,\mathbf{I} \tag{16}$$

where $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$ is the temperature-dependent coefficient of thermal expansion (CTE) and $\Delta T = T - T_0$ is the temperature rise from the stress-free reference state. The thermal strain tensor is $\boldsymbol{\varepsilon}_{\text{th}} = \alpha\,\Delta T\,\mathbf{I}$.

Equilibrium equation. The static momentum balance (in the absence of body forces other than thermal loading) is

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \qquad \text{in } \Omega \tag{17}$$

which, combined with Eqs. (16) and the strain–displacement relation $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$, yields the displacement field $\mathbf{u}$ and the von Mises stress σ_{vM} used as the primary stress output.

F.3. Heat-pipe thermal–hydraulic model

Each sodium heat pipe is modelled with one-dimensional transient conservation equations for the vapour core and the porous wick. The model captures evaporation, vapour transport, condensation and capillary return, producing the steady-state wall temperature for a given input power [9].

Vapour continuity.

$$\frac{\partial \rho_v}{\partial t} + \frac{\partial (\rho_v \, u_v)}{\partial z} = \frac{\dot{m}_v}{D_v} \tag{18}$$

where ρ_v, u_v are the vapour density and axial velocity, z the axial coordinate, $\dot{m}_v$ the mass source from evaporation and D_v the vapour-core diameter.

Liquid continuity in the wick.

$$\frac{\partial (\phi \, S \, \rho_l)}{\partial t} + \frac{\partial (\phi \, S \, \rho_l \, u_l)}{\partial z} = \frac{\dot{m}_l}{D_w} \tag{19}$$

where ϕ is the wick porosity, S the liquid fraction, ρ_l and u_l the liquid density and velocity, $\dot{m}_l$ the condensation mass source.

Dusty-gas vapour transport. To handle the transition between free-molecular and continuum flow regimes inside the vapour core, a second-order dusty-gas model [13] is employed:

$$\frac{\partial (\rho_v \, u_v)}{\partial z} + \frac{\partial}{\partial z}\Big(\rho_v \, u_v^2 + \rho_v \, R_g \, T\Big) = -F_{\text{wall}} \tag{20}$$

where R_g is the specific gas constant and F_{wall} represents the axial flow-diffusion coefficient including wall–molecule collisions.

Darcy's law for wick flow. Liquid flow through the porous wick is modelled by Darcy's law:

$$u_l = -\frac{\kappa}{\mu_l} \frac{\partial P_l}{\partial z} \tag{21}$$

where κ is the permeability of the wire-mesh wick:

$$\kappa = \frac{d_w^2 \, \phi^3}{122 \, (1 - \phi)^2} \tag{22}$$

with d_w the wire diameter. The capillary pressure is $P_c = 2\gamma \cos\theta / r_p$, where γ is the surface tension and r_p the effective pore radius.

Power–temperature database. Because the full transient heat-pipe simulation is expensive, a pre-computed database mapping input power to steady-state wall temperature is used during coupling iterations (420 data points, 1000–8000 W in 100 W steps). Linear extrapolation is applied for out-of-range powers.

F.4. Fuel temperature model

The fuel temperature module computes fuel centreline and average temperatures from the monolith temperature field and the fission power distribution.

Gas-gap conductance. The fuel–monolith gap is treated as a thin helium layer. The helium thermal conductivity is [14]:

$$k_{\mathrm{He}}(T) = A\,T^{B} \tag{23}$$

where A and B are empirical constants ($B = 0.645$ for helium). The temperature drop across the gap is

$$T_{\text{fuel,outer}} - T_{\text{gap,outer}} = \frac{q'}{2\pi\,k_{\mathrm{He}}}\,\ln\!\left(\frac{d_{\text{gap}}}{d_{\text{fuel}}}\right) \tag{24}$$

where q' is the linear power density, d_{gap} the gap outer diameter and d_{fuel} the fuel diameter.

Fuel centreline temperature. For a cylindrical fuel rod with internal heat generation and constant conductivity k_f, the centreline-to-surface temperature difference is

$$T_{\text{centre}} - T_{\text{fuel,outer}} = \frac{q'}{4\pi\,k_f} = \frac{\bar{\dot{q}}\,r_f^2}{4\,k_f} \tag{25}$$

where $\bar{\dot{q}}$ is the average volumetric heat generation rate and r_f the fuel radius. The fuel average temperature is obtained by radial integration:

$$T_{\text{avg}} - T_{\text{fuel,outer}} = \frac{\bar{\dot{q}}\,r_f^2}{8\,k_f} \tag{26}$$

F.5. Picard coupling iteration

The four modules are coupled via Picard (fixed-point) iteration, which sequentially decouples the neutronics, thermal, mechanical and heat-pipe sub-problems. The iteration proceeds as follows:

1. **Neutronics:** OpenMC computes k_{eff}, fission power distribution and key neutronic parameters.
2. **Power mapping:** per-fuel heating tallies are normalised to the rated reactor power to obtain volumetric heat generation rates and surface heat fluxes.
3. **Heat-pipe wall temperature:** the heat-pipe module (or database) converts each pipe's input power to a steady-state wall temperature.
4. **Thermo-elasticity:** FEniCS solves the heat equation (Eq. 13) and the thermoelastic equations (Eqs. 16) on the monolith mesh, yielding temperature, displacement and stress fields. Heat-pipe input powers are updated from the new temperature field.
5. **Inner convergence:** steps 3–4 are repeated until max $|\Delta T_{\text{HP}}| < 1$ K between successive iterations,

with an under-relaxation factor of 0.1 applied to the heat-pipe wall temperatures.

6. **Fuel temperatures:** the fuel-temperature module computes centreline and average temperatures from the converged monolith temperature field.

7. **Geometry update:** displacement-driven coordinate updates for fuel rods and heat pipes are fed back to the OpenMC geometry.

8. **Outer convergence:** material temperatures, densities and coordinates are updated in OpenMC; neutron transport is re-run. The outer loop continues until physical quantities converge or a maximum iteration count is reached.

The neutronics statistical uncertainty is controlled to $< 10\,\text{pcm}$ (1σ) on k_{eff} and $< 0.19\%$ maximum standard deviation on the power distribution. A single coupled solve requires approximately 2266 s on 96 cores (AMD EPYC 9654), with neutronics accounting for $\sim$67% of the total time (see Table S6).

F.6. MegaPower reactor model

The coupled framework is applied to the MegaPower reactor, a $5\,\text{MW}_{\text{th}}$ heat-pipe-cooled microreactor designed by Los Alamos National Laboratory [15]. Key design parameters:

- **Monolith:** SS316 stainless steel; hexagonal lattice containing fuel rods and heat pipes.
- **Fuel:** 19.75 % enriched UO_2 rods ($r_f = 0.71$ cm), 10 rows from inner core to outer core.
- **Heat pipes:** sodium heat pipes ($r_{\text{HP}} = 0.98$ cm), evaporator length 1500 mm, adiabatic section 300 mm, condenser 2200 mm.
- **Reflector:** Al_2O_3 radial reflector with six rotating B_4C control drums for reactivity control.
- **Symmetry:** a 1/6-core sector is modelled with reflective boundary conditions to reduce computational cost.

The FEniCS mesh resolves the monolith cross-section with fuel-rod holes (small) and heat-pipe holes (large). Plane-strain conditions are assumed for the two-dimensional thermo-elastic calculation; a simplified three-dimensional mesh is used for validation.

3.7 Note G: Physics-prior data efficiency

To quantify whether the physics-prior monotonicity constraint improves sample efficiency, we retrain both the unconstrained baseline (ELBO only) and the physics-regularized surrogate (ELBO + monotonicity loss) at 20%, 40% and 60% of the training set (467, 935 and 1403 samples respectively), using the same fixed split, Optuna search (15 trials) and evaluation protocol. All models are evaluated on the full held-out test set ($n = 501$).

At 20% training data (467 samples), the physics-prior model achieves a $+0.009$ advantage in stress R^2 (0.937 vs. 0.928) and a $+0.005$ advantage in wall_2 R^2 (0.991 vs. 0.986), both outputs for which monotonicity constraints are directly imposed. Conversely, k_{eff}—which has no physics-prior pair—shows a slight deficit (0.804 vs. 0.820). At 40% and 60%, the baseline catches up in stress accuracy, and the two models become comparable. This pattern is consistent with the inductive-bias interpretation: the mono-

Table 12: **Table S13 | Per-output R^2 at reduced training fractions.**

Output / fraction	20% base	20% phy	40% base	40% phy	60% base	60% phy	100% base	100% phy
Stress	0.928	0.937	0.939	0.932	0.946	0.939	0.942	0.944
k_{eff}	0.820	0.804	0.822	0.826	0.833	0.836	0.849	0.849
Max fuel temp	0.608	0.600	0.615	0.613	0.627	0.622	—	—
Wall_2	0.986	0.991	0.989	0.989	0.992	0.990	—	—
R^2_{mean} (all 15)	0.744	0.742	0.753	0.749	0.767	0.760	—	—

tonicity prior supplies structural information that partially compensates for missing training samples, but its marginal value diminishes as data accumulate.

The small-sample experiment uses a single random seed for subsampling. Multi-seed repetition would reduce sampling noise but does not change the qualitative conclusion: the physics-prior advantage is concentrated at the lowest data fractions and for constrained outputs.

3.8 Note H: Heteroscedastic vs. homoscedastic noise modelling

To assess whether input-dependent (heteroscedastic) aleatoric noise modelling improves uncertainty quantification, we train an otherwise identical BNN with a single shared log-variance parameter (homoscedastic ablation) and compare it against the reference heteroscedastic surrogate on the same fixed split ($n_{\text{train}} = 2339$, $n_{\text{test}} = 501$).

Table 13: **Table S14 | Heteroscedastic vs. homoscedastic ablation** (test set).

Metric	Hetero (ref)	Homo	Δ (%)
Stress R^2	0.9418	0.9402	-0.2%
k_{eff} R^2	0.8445	0.8327	-1.4%
Stress MPIW (MPa)	40.2	45.2	$+12.4\%$
Stress CRPS	4.424	4.683	$+5.9\%$
PICP	99.0%	99.2%	$+0.2$ pp

Point accuracy (R^2) is virtually unchanged, confirming that the heteroscedastic head does not harm the mean prediction. However, the homoscedastic model produces 12.4% wider predictive intervals for stress and 5.9% higher CRPS, because a single variance must cover both low-uncertainty and high-uncertainty input regions. In the heteroscedastic model, aleatoric standard deviation varies from ≈ 3.1 MPa in low-stress regions to ≈ 8.7 MPa at high stress, reflecting genuine input-dependent noise structure in the coupled simulation. The homoscedastic model, forced to use a single noise scale, inflates intervals uniformly and degrades probabilistic calibration.

Conclusion: heteroscedastic noise modelling is retained in the main-text surrogate because it delivers tighter, better-calibrated intervals at no accuracy cost.

3.9 Note I: Multi-fidelity hybrid architecture ablation

We evaluate whether a multi-fidelity (MF) hybrid BNN architecture—which learns both a residual correction between uncoupled and coupled outputs and a direct mapping for coupled-only outputs—improves over the single-fidelity reference surrogate. The MF-hybrid model uses a two-head architecture: the first

head predicts uncoupled-pass outputs, and the second predicts coupled residuals plus coupled-only outputs directly. Both heads share a common input encoder.

Table 14: **Table S15 | Single-fidelity reference vs. MF-hybrid** (test set, $n = 501$).

Metric	Reference	MF-hybrid	Δ
Stress R^2	0.9418	0.9417	-0.001
k_{eff} R^2	0.8445	0.8110	-0.034
Stress MPIW (MPa)	40.2	47.8	$+18.9\%$
Stress CRPS	4.424	4.772	$+7.9\%$
Speed (s/sample)	0.0158	0.0534	$3.4\times$ slower
HF speedup	143,237$\times$	42,467$\times$	—

At 3418 training samples, the MF-hybrid architecture offers no improvement in point accuracy (stress R^2 identical; k_{eff} R^2 drops by 0.034) and produces wider uncertainty intervals ($+19\%$ stress MPIW). The additional architectural complexity also increases per-sample inference time by $3.4\times$, though absolute speed remains $>42{,}000\times$ faster than coupled HF simulation.

Table 15: **Table S16 | Small-sample data efficiency** (stress R^2).

Training fraction	n_{train}	Reference	Phy-mono	MF-hybrid
20%	467	0.9280	0.9373	0.9108
40%	935	0.9390	0.9315	0.9349
60%	1403	0.9455	0.9391	0.9385
100%	2339	0.9418	0.9440	0.9417

The MF-hybrid was hypothesised to show an advantage under data scarcity, where the uncoupled-pass structure could provide an inductive bias. However, at all training fractions (20%–100%) the reference surrogate matches or exceeds MF-hybrid accuracy. The physics-prior model (phy-mono) shows a modest advantage at 20% training data (stress R^2 0.937 vs. 0.928, $\Delta = +0.009$), but the gap narrows at higher fractions. This pattern suggests that the monotonicity prior provides a mild robustness floor at very low sample sizes but does not confer a sustained data-efficiency advantage.

Conclusion: the MF-hybrid architecture is not adopted for the main-text results. It adds architectural and computational overhead without measurable benefit on this dataset. MF strategies may become relevant when high-fidelity coupled samples are scarce ($n < 500$) and abundant uncoupled data are available—a regime not explored in this study.

3.10 Note J: Posterior-mean HF re-evaluation

To verify that the surrogate-side posterior summaries in Section 2.4 are consistent with the coupled solver, we re-run the full OpenMC–FEniCS pipeline at three posterior parameter points—the posterior mean, the 5th-percentile vector and the 95th-percentile vector—for each of the 18 benchmark cases, yielding 54 independent HF evaluations. The four calibrated parameters ($E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}) are set to their MCMC posterior summary values; the remaining four parameters are fixed at the test-split true values. All runs use the same solver configuration as the training-data generation campaign (two-iteration Picard coupling, SS316_scale $= 0.01$).

The stress error at the posterior mean is comparable to the surrogate's test-set MAE, confirming that the

Table 16: **Table S17 | Post-mean HF accuracy** (18 cases).

Output	MAE	RMSE	rel MAE
k_{eff} (coupled)	0.0005	0.0007	0.05%
Max fuel temperature (K)	4.79	5.67	0.45%
Max monolith temp (K)	5.34	6.33	0.52%
Max global stress (MPa)	5.65	6.44	4.52%
HP wall thickness (cm)	0.020	0.025	0.07%

MCMC-inferred parameter regions remain within the surrogate's accurate prediction domain. Stress MAE by category: low 6.65 MPa, near-threshold 6.65 MPa, high 3.67 MPa. The high-stress cases show smaller posterior-mean error, consistent with the posterior contracting toward the correct high-stress parameter region.

Table 17: **Table S18 | [5%, 95%] HF envelope coverage.**

Output	Coverage
k_{eff} (coupled)	17/18 = 0.944
Max fuel temperature (K)	13/18 = 0.722
Max monolith temp (K)	13/18 = 0.722
Max global stress (MPa)	18/18 = 1.000
HP wall thickness (cm)	17/18 = 0.944

Stress achieves full coverage: for all 18 cases the true stress falls between the HF solutions at the 5th- and 95th-percentile posterior parameters. The temperature outputs show lower coverage (13/18) because the posterior parameter perturbation at extreme percentiles produces larger temperature shifts than stress shifts, and several true values lie just outside the [lo5, hi95] HF band. This reflects the posterior width in temperature space rather than a calibration failure.

Table 18: **Table S19 | Per-case post-mean stress and k_{eff}.**

Case	Cat.	σ_{true}	σ_{HF}	$\lvert\Delta\sigma\rvert$	$k_{\text{eff,true}}$	$k_{\text{eff,HF}}$	$\lvert\Delta k\rvert$
0	low	110.57	119.95	9.38	1.10457	1.10422	0.00035
1	low	118.66	121.02	2.36	1.10377	1.10412	0.00035
2	low	110.65	115.54	4.89	1.10225	1.10398	0.00174
3	low	103.67	113.10	9.43	1.10448	1.10424	0.00024
4	low	114.50	121.89	7.39	1.10387	1.10347	0.00039
5	low	102.00	108.43	6.43	1.10443	1.10397	0.00045
6	near	127.60	134.20	6.60	1.10389	1.10386	0.00003
7	near	127.92	131.18	3.26	1.10476	1.10383	0.00093
8	near	126.98	137.45	10.47	1.10400	1.10391	0.00009
9	near	123.51	131.90	8.39	1.10302	1.10413	0.00111
10	near	128.92	132.76	3.84	1.10294	1.10415	0.00121
11	near	129.29	136.61	7.32	1.10437	1.10390	0.00047
12	high	169.54	174.48	4.94	1.10321	1.10323	0.00002
13	high	176.02	174.56	1.46	1.10417	1.10345	0.00073
14	high	193.40	193.40	0.00	1.10328	1.10314	0.00014
15	high	163.37	162.83	0.54	1.10339	1.10359	0.00019
16	high	142.79	150.76	7.97	1.10448	1.10374	0.00074
17	high	197.64	190.54	7.10	1.10318	1.10353	0.00035

The 54 HF evaluations ran on a single AMD EPYC 9654 96-core node (566 GB RAM, 2× NVIDIA RTX

5090). Mean wall-clock time per case: 2357 s (median 2328 s, $\sigma = 297$ s, range 1768–3022 s). Total elapsed time for the batch: approximately 35 hours with sequential execution and automatic skip-on-completion restart.

Supplementary References

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